

# Ahmad Ghasemi

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🏠 University of Massachusetts, Amherst, MA | Work authorization: U.S. Permanent Resident

## RESEARCH PROFILE

I am a graph machine learning researcher developing theoretically grounded methods for structured decision-making problems. My work studies when local graph representations and message-passing architectures can generalize across problem instances, and how these insights can support reliable ML-guided optimization, online decision-making, and learning under uncertainty.

**Core research areas:** graph neural networks; statistical learning theory; sample complexity; locality and message passing; structure-aware generalization; learning-augmented algorithms; optimization-aware learning; adversarial robustness; uncertainty-aware and efficient inference.

**Research leadership:** Co-PI and key personnel on externally funded and submitted NSF projects involving graph-based modeling, trustworthy neural learning, multi-agent decision systems, and learning-and-control evaluation pipelines.

## RESEARCH FUNDING AND PROPOSAL LEADERSHIP

|              |                                                                                                                                                                                                                                                                                                                                                                                                           |
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| Active       | <b>NSF CPS Small, Co-PI, AERIAL:</b> AI-Embedded Responsive Intelligent Agents with Trajectory-Induced Digital Twin Learning, CNS-2528914, \$599,961, 08/2025–07/2028. <i>PI:</i> H. Pishro-Nik. <i>Role:</i> Lead investigator for graph-based modeling, learning-and-control, budget-aware neural inference, robustness evaluation, and performance metrics for multi-agent networked decision systems. |
| Under Review | <b>NSF CyberAI SFS, Senior Personnel, CyberAI Innovation:</b> An Educational Security Testbed for Networked AI Agents, \$499,680. <i>PI:</i> H. Pishro-Nik, <i>Co-PI:</i> A. Houmansadr. <i>Submitted:</i> Apr. 2026. <i>Role:</i> Contributed to the technical vision, AI-agent testbed design, uncertainty-aware reasoning activities, and research-to-education integration.                           |

## RESEARCH PROGRAM AND SELECTED CONTRIBUTIONS

- **Graph neural network theory and topology-dependent generalization.** Established minimax lower bounds and structural regimes for message-passing graph neural networks, clarifying when learnability is governed by graph topology, label availability, feature dimension, and spectral-homophily structure. This work connects statistical learning theory with the design and evaluation of graph-based learning systems.
- **Learning-augmented decision systems under uncertainty.** Developed graph learning and neural inference methods for decision-making in online, networked, and resource-constrained systems, where models must operate under uncertainty, limited information, latency constraints, and changing system conditions. This work includes graph-based resource allocation, UAV/network coordination, and evaluation pipelines for learned decision policies.
- **Robust graph learning and adversarial evaluation.** Developed adversarial evaluation and robustness methods for graph-based resource allocation, with applications to wireless and cyber-physical decision systems. This work identifies failure modes of learned graph policies under adversarial perturbations, imperfect system information, and distribution shift.
- **Resource-aware and constrained graph learning.** Designed compact, low-rank, and constrained neural models for inference under memory, latency, energy, and communication limits. This includes tiny graph neural networks for radio resource management and constrained low-rank training methods for graph neural networks.
- **Efficient reasoning and confidence-guided inference.** Contributed to work on confidence-guided early stopping for efficient inference, connecting uncertainty-aware model behavior with computational efficiency in large-model reasoning systems.

## PROFESSIONAL APPOINTMENTS

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| <b>Postdoctoral Research Fellow</b> , University of Massachusetts, Amherst, MA<br><i>Supervisor:</i> Hossein Pishro-Nik.<br><i>Focus:</i> Graph neural networks and structured learning for decision-making systems, including GNN sample complexity, topology-dependent generalization, robustness under distribution shift/adversarial perturbations, and efficient inference. | <b>05/2024 - Present</b> |
| <b>Adjunct Faculty (Instructor of Record / Co-Instructor)</b> , University of Massachusetts, Amherst, MA<br><i>Courses:</i> Machine Learning for Engineers, Foundations of Generative Models, Digital Signal Processing, and Digital Image Processing (graduate/undergraduate; project-based delivery).                                                                          | <b>09/2023 - Present</b> |

## EDUCATION

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| <b>Ph.D., Data Science</b> , Worcester Polytechnic Institute (WPI), Worcester, MA<br><i>Dissertation:</i> Intelligent and Resilient Resource Allocation in mmWave Wireless Communications<br><i>Research areas:</i> graph neural networks, graph-based optimization, learning-based resource allocation, adversarial robustness, and decision-making under uncertainty.<br><i>Advisor:</i> Reza Zekavat | <b>2023</b> |
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| <b>M.Sc., Electrical and Computer Engineering</b> , Shiraz University, Shiraz                    | <b>2012</b> |
| <i>Dissertation:</i> Machine Learning-Based Spectrum Allocation in Cognitive Radio Networks      |             |
| <i>Advisors:</i> Mohammad Ali Masnadi-Shirazi; Mehrzad Biguesh                                   |             |
| <b>B.S., Electrical and Computer Engineering</b> , University of Sistan and Baluchestan, Zahedan | <b>2007</b> |

## PUBLICATIONS

\* indicates co-first authorship.

### Peer-Reviewed Conference Proceedings

- [C1] **A. Ghasemi** and H. Pishro-Nik, "Minimax Sample Complexity of Graph Neural Networks: Lower Bounds and Structural Effects," The Fourteenth International Conference on Learning Representations (ICLR) 2026. OpenReview PDF.
- [C2] M. Malekzadeh, **A. Ghasemi**, and H. Pishro-Nik, "Polar-Grid Graph Attention for Finite-Horizon Density-Aware UAV Coverage," IEEE Vehicular Technology Conference, Boston, MA, 2026.
- [C3] M. Malekzadeh\*, **A. Ghasemi**\*, and H. Pishro-Nik, "Constant-Speed Trajectory Processes for Uniform Coverage in UAV Networks," 59th Annual Conference on Information Science & Systems (CISS), Baltimore, Maryland, USA, 2025.
- [C4] **A. Ghasemi** and H. Pishro-Nik, "Tiny Graph Neural Networks for Radio Resource Management," in *Proceedings of tinyML Research Symposium (tinyML Research Symposium'24)*, ACM, Burlingame, CA, USA, 2024, pp. 1–7.
- [C5] **A. Ghasemi**, M. Moradikia, R. Zekavat and H. Pishro-Nik, "Adversarial Attacks Targeting Point-to-Point Wireless Networks," 2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring), Singapore, Singapore, 2024, pp. 1–6.
- [C6] **A. Ghasemi**, E. Zeraatkar, M. Moradikia, and R. Zekavat, "Adversarial Attacks on Resource Management in P2P Wireless Communications," 2023 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Aveiro, Portugal, 2023, pp. 148–153.
- [C7] **A. Ghasemi** and R. Zekavat, "On Eigenvalue Distribution of Imperfect CSI in mmWave Communications," 2022 IEEE USNC-URSI Radio Science Meeting (Joint with AP-S Symposium), Denver, CO, USA, 2022, pp. 56–57. (*Travel Award*)
- [C8] **A. Ghasemi** and R. Zekavat, "Joint Hybrid Beamforming and Dynamic Antenna Clustering for Massive MIMO," 2020 29th Wireless and Optical Communications Conference (WOCC), Newark, NJ, USA, 2020, pp. 1–6. (*Best Paper Award*)
- [C9] **A. Ghasemi**, A. F. Jahromi, M. A. Masnadi-Shirazi, M. Biguesh and F. Ghasemi, "Spectrum Allocation Based on Artificial Bee Colony in Cognitive Radio Networks," 6th International Symposium on Telecommunications (IST), Tehran, Iran, 2012, pp. 182–187.
- [C10] **A. Ghasemi**, A. F. Jahromi, M. A. Masnadi-Shirazi, M. Biguesh and F. Ghasemi, "Spectrum Allocation with Control of Interference Based on Differential Evolution Algorithm Between Cognitive Radio Users," Proc. 20th Iranian Conf. Elect. Eng. (ICEE), 2012.
- [C11] Z. Iqbal, S. Nooshabadi, K. Jadi and **A. Ghasemi**, "Sensor Cooperation and Decision Fusion to Improve Detection in Cognitive Radio Spectrum Sensing," 9th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, 2018, pp. 276–281.

### Peer-Reviewed Journal Articles

- [J1] M. Malekzadeh\*, **A. Ghasemi**\*, and H. Pishro-Nik, "Robust UAV Trajectory Design for Non-Uniform Coverage," IEEE Communications Letters, vol. 30, pp. 188–192, 2026.
- [J2] **A. Ghasemi**, E. Zeraatkar, M. Moradikia, and R. Zekavat, "Adversarial Attacks on Graph Neural Networks-Based Spatial Resource Management in P2P Wireless Communications," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 8847–8863, June 2024.
- [J3] **A. Ghasemi** and S. A. Zekavat, "Low-Cost mmWave MIMO Multi-Streaming via Bi-Clustering, Graph Coloring, and Hybrid Beamforming," *IEEE Transactions on Wireless Communications*, vol. 20, no. 7, pp. 4113–4127, July 2021.
- [J4] **A. Ghasemi**, M. Masnadi-Shirazi, M. Biguesh, and F. Qassemi, "Channel Assignment Based on Bee Algorithms in Multi-hop Cognitive Radio Networks," *IET Communications*, 8(13), 2356–2365, 2014.
- [J5] R. Zekavat, R. M. Buehrer, G. D. Durgin, L. Lovisolo, Z. Wang, T. Goh, and **A. Ghasemi**, "An Overview on Position Location: Past, Present, Future," *International Journal of Wireless Information Networks*, 28, 45–76 (2021).

### Peer-Reviewed Workshop Papers

- [W1] E. Aghazade, **A. Ghasemi**, H. Beyhaghi, and H. Pishro-Nik, "CGES: Confidence-Guided Early Stopping for Efficient Inference," Efficient Reasoning Workshop, NeurIPS 2025. ArXiv.

### Preprints and Manuscripts Under Review

- [P1] S. Amini\*, **A. Ghasemi**\*, and H. Pishro-Nik, "Constrained Optimization for Low-Rank Training of Graph Neural Networks," Under review (*IEEE Transactions on Signal Processing*). Preprint available upon request.
- [P2] E. Aghazade, M. Malekzadeh, **A. Ghasemi**, and H. Pishro-Nik, "Ergodic Trajectory Design by Learned Pushforward Maps: Provable Coverage via Conditional Flow Matching," Under review (*NeurIPS 2026*). ArXiv.

[P3] **A. Ghasemi**, M. Malekzadeh, E. Aghazade, and H. Pishro-Nik, “Tiny Flow Matching: Constrained Low-Rank OT-CFM for Deployable Ergodic UAV Coverage,” Under review (*IEEE International Workshop on Machine Learning for Signal Processing*). Preprint available upon request.

## Book

[B1] F. Ghasemi, **A. Ghasemi**, “Numerical Methods: Basics and coding in MATLAB,” Zolal-e-Sabz Press, 2020.

## RESEARCH MENTORSHIP

**Research mentor / co-mentor:** 3 Ph.D. students, 1 M.Sc. student, and 5 undergraduate researchers at UMass Amherst.

**Mentoring areas:** graph neural networks, UAV trajectory learning, efficient LLM inference, neural architecture search, generative AI, and AI-tutor systems.

**Selected outcomes:** IEEE journal and conference publications; NeurIPS workshop acceptance; EMNLP/NeurIPS/major-venue submissions; honors thesis direction; reproducible evaluation harnesses.

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| Spring 2025 – Present | <b>Ehsan Aghazadeh</b> , Ph.D. student, Computer Science<br><i>Focus:</i> Efficient and accurate LLM test-time scaling.<br><i>Outcome:</i> NeurIPS workshop acceptance and EMNLP submission.                                                                                      | UMass Amherst |
| 2024–Present          | <b>Masoud Malekzadeh</b> , Ph.D. student, Electrical and Computer Engineering<br><i>Focus:</i> Graph-based and learning-based unmanned aerial vehicle (UAV) trajectory design.<br><i>Outcome:</i> IEEE journal and conference publications.                                       | UMass Amherst |
| Fall 2025 – Present   | <b>Giap Hoang Nguyen</b> , UG student, Mathematics and Computer Science<br><i>Focus:</i> UAV orchestration using graph neural networks.<br><i>Outcome:</i> Project completed; manuscript/report in preparation.                                                                   | UMass Amherst |
| Fall 2025             | <b>Thuyen Pham</b> , UG student, Mathematics and Computer Science<br><i>Focus:</i> Automated imaging for butterfly identification.<br><i>Outcome:</i> Research direction adopted as Honors thesis.                                                                                | UMass Amherst |
| Summer 2025           | <b>Huy Gia Cao, Jake Reid, Mykaala Firdaus, Ritik Shah</b> , undergraduate researchers, Electrical and Computer Engineering<br><i>Focus:</i> LLM fine-tuning and AI-tutor prototyping.<br><i>Outcome:</i> Fine-tuned LLM; curated training dataset; completed AI-Tutor prototype. | UMass Amherst |

## TALKS

|            |                                                                                                                                                   |                |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| March 2026 | <b>University of Texas, Tyler</b><br><i>Trustworthy and Resource-Aware AI: From Graph Learning Theory to Deployment</i>                           | Invited        |
| March 2026 | <b>Florida Polytechnic University, Lakeland</b><br><i>Building Reliable and Efficient ML for Real-World Decision-Making</i>                       | Invited        |
| March 2026 | <b>California State University, Long Beach</b><br><i>Trustworthy and Secure Edge AI under Real-World Constraints</i>                              | Invited        |
| May 2024   | <b>TinyML Research Symposium</b><br><i>Tiny Graph Neural Networks for Radio Resource Management</i>                                               | Burlingame, CA |
| May 2023   | <b>NEWSDR: New England Workshop on Software-Defined Radio</b><br><i>Adversarial Attacks on Graph Neural Network-based Wireless Communications</i> | Worcester, MA  |

## PROFESSIONAL SERVICE

- NSF CISE and TTP-T Panels Reviewer, 2026.
- Reviewer: *ICLR*; *IEEE Transactions on Wireless Communications*; *IEEE Transactions on Vehicular Technology*; *IEEE Transactions on Communications*; *IEEE Transactions on Information Forensics and Security*; *EURASIP Journal on Wireless Communications and Networking*.
- Organizing Chair, IEEE WiSEE 2020.
- Curriculum Committee Member, UMass iCons track “AI and the Future of Work,” 2024–present.

## TEACHING EXPERIENCE

**Teaching areas:** machine learning, generative models, probability/statistics, data science foundations, graph neural networks, signal processing, image processing, and trustworthy/efficient AI.

**Teaching effectiveness:** Recent UMass Amherst instructor effectiveness **4.7/5.0**; well prepared **5.0/5.0**; used class time well **5.0/5.0**; showed interest in helping students learn **5.0/5.0**.

#### Instructor of record / Co-Instructor

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|---------------------------------------|---------------------------------------------------------------------------------------------------|---------------|
| Fall 2025                             | <b>ECE690: Foundations of Generative Models (Graduate)</b><br><i>Co-Instructor</i>                | UMass Amherst |
| Fall 2025                             | <b>ECE150: Better Decisions by Human &amp; AI (Undergraduate)</b><br><i>Instructor</i>            | UMass Amherst |
| Spring 2025, Summer 2024, Spring 2024 | <b>ECE601: Machine Learning for Engineers (Graduate)</b><br><i>Instructor / Co-Instructor</i>     | UMass Amherst |
| Fall 2024                             | <b>ECE565: Digital Signal Processing &amp; Representation (Graduate)</b><br><i>Instructor</i>     | UMass Amherst |
| Summer 2024                           | <b>ECE579: Math Tools for Data Science &amp; Machine Learning (Graduate)</b><br><i>Instructor</i> | UMass Amherst |
| Fall 2023                             | <b>ECE566: Digital Image Processing (Graduate)</b><br><i>Instructor</i>                           | UMass Amherst |

#### Teaching Assistant

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|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|-----|
| Spring 2022, Fall 2021, Fall 2020 | <b>DS517: Math Foundations for Data Science (Graduate)</b><br><i>Graduate Teaching Assistant</i>                   | WPI |
| Spring 2021                       | <b>DS541: Deep Learning (Graduate)</b><br><i>Graduate Teaching Assistant</i>                                       | WPI |
| Spring 2020                       | <b>DS2010: Data Science II: Modeling &amp; Data Analysis (Undergraduate)</b><br><i>Graduate Teaching Assistant</i> | WPI |
| Fall 2019                         | <b>ECE2312: Signal and System Analysis (Undergraduate)</b><br><i>Graduate Teaching Assistant</i>                   | WPI |

#### Course & Curriculum Development

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|--------------|--------------------------------------------------------------------------------------------------------------------|---------------|
| 2024–Present | <b>Curriculum Development for New iCons Track</b><br><i>AI and the Future of Work, Curriculum Committee Member</i> | UMass Amherst |
| Fall 2025    | <b>Developed New Graduate Course</b><br><i>ECE690: Foundations of Generative Models</i>                            | UMass Amherst |
| Spring 2024  | <b>Developed New Graduate Course</b><br><i>ECE601: Machine Learning for Engineers</i>                              | UMass Amherst |

## INDUSTRY AND APPLIED RESEARCH EXPERIENCE

**Industry-Sponsored Research Intern**, Ford Motor Company / Wireless Positioning Lab, Michigan Tech **Summer 2019**

- Built reproducible perception and positioning experiments for autonomous-driving scenarios under sensing uncertainty and environmental perturbations.

**Data Scientist**, Hormozgan Electricity Distribution Co., Bandar Abbas **2015–2018**

- Developed forecasting, anomaly detection, and decision-support workflows for power-distribution reliability.

## HONORS AND AWARDS

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|------------------------------------------------------------------------------------------------------------|-------------|
| <b>Charles Kao Best Paper Award</b> , 29th Wireless and Optical Communications Conference, Newark, NJ, USA | <b>2020</b> |
| <b>Travel Award</b> , School of Arts & Sciences, Worcester Polytechnic Institute, Worcester, MA, USA       | <b>2022</b> |
| <b>Nominee, TA of the Year Award</b> , Worcester Polytechnic Institute, Worcester, MA, USA                 | <b>2022</b> |